Ajinkya Pawale

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Experience

AI Engineer — Chicago, IL
Elevance Health — 01/2023 - Present

- Enhanced and optimized existing data retrieval setups for Inpatient Claims using **PySpark's** distributed framework, achieving a **75% reduction** in data extraction time, and enriching machine learning model inputs with **20+ derived features** using **Pytorch**.
- Spearheaded the automation of data pipelines in **Airflow**, integrating **ETL, modeling, and post-processing** components for the Inpatient Claims project, leveraging **ML tooling** and saving almost 4 hours of manual work during every production run.
- Led end-to-end GenAI solution for automated call summarization using **OpenAI GPT models**, architecting **Langchain**-based prompt orchestration framework that achieved **55%** faster response times; drove adoption across **350+ agents**, reducing manual wrap-up time by **5 minutes/call** and improving operational efficiency
- Augmented the call wrap-up API with **RAG pipelines** retrieving customer context from internal knowledge bases and **autonomous agent workflows** for validation, achieving **85%** higher summary quality scores and cutting post-call correction time by **two-thirds**.
- Collaborated with Data Science teams on feature enhancements, performing data curation and **full stack ML development** to refine call summary generation within Call Center projects. Detailed prompts improved caller details and metadata capture accuracy by **10%**, enhancing agent follow-up response rates.
- Architected and deployed a **multi-agent GPT-4.1-based** document compliance system that automated style guide enforcement for **50+** L&D team members, leveraging specialized agents (text/tone, formatting, image analysis, layout) with parallel processing to overcome LLM context window limitations and reduce manual review time by **80%** for large-scale training materials.
- Designed a **Kinesis Consumer Application** for processing and post-processing live call transcript data from an event bus, leveraging the **Kinesis Streams** architecture alongside **Redis cache** to enhance data retrieval speed by 40%.
- Established **20+ crucial alerts and dashboards** in **Datadog and Splunk** to monitor system health, accuracy and performance, reducing incident response time by a significant amount.
- Orchestrated a **pytest** framework covering 50+ test scenarios that streamlined automated testing processes within the development cycle of APIs, directly **reducing** manual testing time by an average of **three hours** per week.
- Executed load and stress tests using **LoadRunner** in collaboration with the performance team, and monitored key performance metrics via Datadog. Coordinated API-based **regression testing** to ensure consistency in performance across feature cycles.

Data Engineer Intern — Chicago, IL
Elevance Health — 06/2022 - 08/2022

- Enhanced a Machine Learning project with **BOHB (Bayesian Optimization with HyperBand)**, boosting model efficiency by **12%**, and contributed to designing ML frameworks and autonomous AI solutions to minimize manual intervention.
- Implemented a robust **data validation** tool that detected **production data drift** by profiling training datasets and validating against **15** specific scoring checkpoints, resulting in nearly **two full days** of saved workload each month.
- Migrated project data from **on premise** cluster to **AWS EMR** cluster, improving processing **speed, scalability, and reliability**. Implemented best practices for data transfer and cloud architecture, ensuring minimal downtime and integrity.

Research Assistant — Bloomington, IN
Indiana University — 06/2021 - 12/2022

- Conducted in-depth **statistical analysis in R, pandas, and scikit-learn**, reducing data dimensions and demonstrating re-identification risks in de-identified data, leading to a **20%** improvement in predictive accuracy.
- Performed statistical analysis on **100,000+ data entries** from six distinct sources using **numpy** to evaluate student engagement in course discussions for a cohort of **800 learners**; utilized **Tableau** visualizations to reduce manual analysis time by more than half.
- Built an **Artificial Neural Network** using TensorFlow to predict student retention from library usage. The Network provided valuable information on student behaviors that correlate with retention, helping to identify at-risk students early.

Data Engineer — Mumbai, India
Course5 Intelligence — 12/2019 - 12/2020

- Spearheaded process optimization efforts through **tailored automation** initiatives on an e-commerce platform, leading to the successful elimination of more than **150 repetitive tasks** with significant improvements in workflow speed.
- Undertook an entire process for systematic **Product Inventory** updates and health checks for the website using **Java and Selenium**, providing end-to-end solutions and possible ways to optimize the process.
- Created a robust analytics framework utilizing **pymongo** to query the newly created **data lake**, which enabled generating actionable insights across **250+ product categories** and improved reporting efficiency by minimizing manual reporting efforts.

Technical Skills

Programming/Scripting: Python, PySpark, R, SQL, Javascript, XML, JSON, HTML5, CSS3, PowerShell
Algorithms: Regression, Classification, Clustering, XGboost, Naye Bayes, SVMs, CNN, RNN, LSTM, MLPs, GRU, GANs, Transformers
Technical Tools: Microsoft Suite, Tableau, PowerBI, VBA, PyTorch, TensorFlow, Pandas, NumPy, Scikit Learn, Hugging Face, Computer Vision, Predictive/Descriptive Analytics, ANOVA, Chi-Square, Functions, Triggers, Key Vault, FastAPI, Pydantic
Databases: MSSQL, PostgreSQL, T-SQL, MySQL, MongoDB, Neo4j, SSMS, SSIS, Hadoop, Hive
Cloud Infrastructure: S3, EC2, Lambda, Kubernetes, ECS, ECR, Elastic Cache, Glue, CloudWatch
Developer Tools: Git, Docker, Postman, Visual Studio, GitLab, Bitbucket, JIRA

Education

Indiana University — Bloomington, IN
Master of Science in Data Science — 01/2021 - 12/2022